D206: Data Cleaning Performance Assessment

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A1:Question or decision

According to Jaladh Singhal (2020, June 29), formulating a research question states the problem you seek to solve by analysis.

The objective of the analysis of the Churn_raw_data.csv is to assess the features that highly impact the customers in the telecommunication industry and cause the customer to switch providers to answer the question of what values are important in customer retention and would highly impact the churn rate.

A2: Required variables

Chan, Jamie (2014), (pg 17) states that Variables are names given to data that we need to store and manipulate in our programs. According to Sharma, Rohit (2024, Jan 24), Data types classify data values into some data categories. The built-in data types in Python include Binary, Boolean, set types, Mapping, sequence, Numeric, and Text Type

The Churn dataset has the following variables:

1. 'Unnamed'

   The Unnamed variable has 10,000 unique values of numerical data of integer data type. The numbers are sequential, but we cannot perform statistical measures on the data since it is a unique identifier. The data type for the unnamed variable is integer (int).

   Example of the unnamed variable data

1
2
3
4
5
6

2. CaseOrder

   The CaseOrder variable has 10,000 unique values of numerical data of integer data type. This is a placeholder variable. The numbers are sequential, but we cannot perform statistical measures on the data since it is a unique identifier. The data type for the CaseOrder variable is integer (int).

   Example of the CaseOrder variable data

CaseOrder
1
2
3
4
5
6

3. Customer_id

   The Customer_id is a unique customer identifier with a minimum length of six and a maximum length of seven. The Customer_id data has alphanumeric characters; hence, the data type for the Customer_id is a string (str).

   Example of the Customer_id variable data

Customer_id
K409198
S120509
K191035
D90850
K662701

4. Interaction

The Interaction variable is a unique identifier for each customer, with thirty-six characters uniquely identifying each object for each customer's transaction. The data is categorical; hence the data type is a string (str).

Example of the Interaction variable data

Interaction
aa90260b-4141-4a24-8e36-b04ce1f4f77b
fb76459f-c047-4a9d-8af9-e0f7d4ac2524
344d114c-3736-4be5-98f7-c72c281e2d35
abfa2b40-2d43-4994-b15a-989b8c79e311
68a861fd-0d20-4e51-a587-8a90407ee574

5. City

The city variable has the customers' city of residence. The data is categorical; hence the data type is a string (str).

Example of the city variable data

City
Point Baker
West Branch
Yamhill
Del Mar
Needville
Fort Valley
Pioneer
Oklahoma City

6. State

   The state variable is the customers' state of residence. The data is categorical; hence the data type is a string (str).

   Example of the State variable data

State
AK
MI
OR
CA
TX
GA

7. Country

   The Country variable has the customers' Country of residence. The data is categorical; hence the data type is a string (str).

   Example of the State variable data

County
Prince of Wales-Hyder
Ogemaw
Yamhill
San Diego
Fort Bend
Peach
Scott

8. Zip

   The Zip variable has the customers' Zip of residence. The Zip data is numerical with whole numbers, but since the Zip data cannot be summed up since it is categorical data, the data type for the Zip variable is string (str).

   Example of the Zip variable data

Zip
99927
48661
97148
92014
77461
31030

9. Lat

   The Lat variable has the GPS latitude coordinate of the customer's residence. The Lat data is numerical with decimals, but we cannot calculate statistical measures on the data since it is categorical data; hence, the data type for the Lat variable is a string (str).

   Example of the Lat variable data

Lat
56.251
44.32893
45.35589
32.96687
29.38012

10. Lng

The Lng variable has the GPS longitude coordinate of the customers' residences. The Lng data is numerical with decimals, but we cannot calculate statistical measures on the data since it is categorical data; hence, the data type for the Lng variable is string (str).

Example of the Lng variable data

Lng
-133.37571
-84.2408
-123.24657
-117.24798
-95.80673
-83.8904

11. Population

The Population variable has the total population within a mile of the customer's residence. The population data is numerical with whole numbers, and we can calculate statistical measures on the data; hence, the data type for the Population variable is integer (int).

Example of the population variable data

Population
38
10446
3735
13863
11352
17701

12. Area

The Area variable describes the zoning of the customer's areas of residence. The data is categorical; hence the data type is a string (str).

Example of the Area variable data.

Area
Urban
Suburban
Rural

13. Timezone

The Timezone variable describes the time zoning of the customer's areas of residence. The data is categorical; hence the data type is a string (str).

Example of the Timezone variable data.

Timezone
America/Sitka
America/Detroit
America/Los_Angeles
America/Chicago
America/New_York
America/Puerto_Rico
America/Denver

14. Job

The Job variable describes the occupation of the customers. The data is categorical data that can be used for classification. The data type is a string (str).

Example of the Job variable data.

Job
Environmental health practitioner
Programmer, multimedia
Chief Financial Officer
Solicitor
Medical illustrator

15. Children

The Children variable describes the number of Children a customer has. The data is numerical data that can be used for classification, and we can perform statistical measures on the data. The data has missing values as represented by 'NA.' The data type is integer (int).

Example of the Children variable data.

Children
0
1
2
3
4
5
6
7
8
9
10
NA

16. Age

The Age variable describes the number of customers. The data is numerical data that can be used for classification as well if grouped into clusters. Also, we can perform statistical measures on the data. The data has missing values as represented by ‘NA.’ The data type is integer (int).

Example of the Age variable data.

Age
68
27
50
48
83
NA
49

17. Education

The Education variable describes the customer's education level. The data is categorical data that can be used for classification. The data type is a string (str).

Example of the Education variable data.

Education
Master's Degree
Regular High School Diploma
Doctorate Degree
No Schooling Completed
Associate's Degree
Bachelor's Degree
Some College, Less than 1 Year
GED or Alternative Credential
Some College, 1 or More Years, No Degree
9th Grade to 12th Grade, No Diploma
Nursery School to 8th Grade
Professional School Degree

18. Employment

The Employment variable describes the customer's employment status. The data is categorical data that can be used for classification. The data type is a string (str).

Example of the Employment variable data.

Employment
Part Time
Retired
Student
Full Time
Unemployed

19. Income

The Income variable describes the number of customer's income. The data is numerical data that can be used for classification as well if grouped into clusters. also, we can perform statistical measures on the data. The data has missing values as represented by 'NA.' The data type is float.

Example of the income variable data.

Income
28561.99
21704.77
NA
18925.23
40074.19
11467.5
26759.64

20. Marital

The Marital variable describes the customer's marital status. The data is categorical data that can be used for classification. The data type is a string (str).

Example of the Marital variable data.

Marital
Widowed
Married
Separated
Never Married
Divorced

21. Gender

The Gender variable describes the customer's self-identification. The data is categorical data that can be used for classification. The data type is a string (str).

Example of the Gender variable data.

Gender
Male
Female
Prefer not to answer

22. Churn

The Churn variable describes if the customer left in the last month. The data type is Boolean and can be used as a target variable. The data type is Boolean (bool).

Example of the Churn variable data.

Churn
No
Yes

23. Outage_sec_perweek

The Outage_sec_perweek variable describes the average number of seconds per week of systems outages in customers' neighborhoods. The data is numerical; hence, we can perform statistical measures on the data. The data type is float.

Example of the Outage_sec_perweek variable data.

Outage_sec_perweek
6.972566093
12.01454108
10.24561565
15.20619309

24. Email

The Email variable shows the number of marketing email correspondences that have been sent to the customer. The data type is integer (int).

Example of the Email variable data.

Email
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23

25. Contacts

The Contacts variable shows the number of times the customer contacted the technical support. The data type is integer (int).

Example of the Contacts variable data.

Contacts
0
1
2
3
4
5
6
7

26. Yearly_equip_failure

The Yearly_equip_failure variable shows the number of times a customer's equipment failed and had to be reset/replaced in the past year.

The data type is integer (int).

Example of the Yearly_equip_failure variable data.

Yearly_equip_failure
0
1
2
3
4
6

27. Techie

The Techie variable represents if the customer is technically inclined. The data has missing values represented by 'NA.' The data type is Boolean (bool).

Example of the Techie variable data.

Techie
No
Yes
NA

28. Contract

The Contract variable represents the contract terms of the customer. The data type is a string (str).

Example of the Contract variable data.

Contract	
One year	
Month-to-month	
Two Year	

29. Port_modem

The Port_modem indicates whether the customer has a portable modem. The data type is Boolean (bool)

Example of the Port_modem variable data.

	Port_modem	
	Yes	
	No	

30. Tablet

The Tablet variable indicates whether the customer owns a tablet such as an iPad, surface, etc. The data type is Boolean (bool)

Example of the Tablet variable data.

Tablet
Yes
No

31. InternetService

The InternetService variable indicates the customer's InternetService provider. The data has missing values represented by 'None.' The data type is a string (str).

Example of the InternetService variable data.

InternetService
Fiber Optic
DSL
None

32. Phone

The Phone variable indicates if the customer has a phone service or not. The data has missing values represented by 'NA.' The data type is Boolean (bool).

Example of the Phone variable data.

Phone
Yes
No
NA

33. Multiple

The Multiple variable indicates if the customer has multiple lines. The data type is Boolean (bool).

Example of the Multiple variable data.

Multiple
No
Yes

34. OnlineSecurity

The OnlineSecurity variable indicates whether the customer has an online Security add-on. The data type is Boolean (bool).

Example of the OnlineSecurity variable data.

OnlineSecurity
Yes
No

35. OnlineBackup

The OnlineBackup variable indicates whether the customer has an online backup add-on. The data type is Boolean (bool).

Example of the Online Backup variable data.

OnlineBackup
Yes
No

36. DeviceProtection

The DeviceProtection variable indicates whether the customer has a device protection add-on. The data type is Boolean (bool).

Example of the Device Protection variable data.

DeviceProtection
No
Yes

37. TechSupport

The TechSupport variable indicates whether the customer has a technical support add-on. The data has missing values represented by 'NA.' The data type is Boolean (bool).

Example of the TechSupport variable data.

TechSupport
No
Yes
NA

38. StreamingTV

The StreamingTV variable indicates whether the customer has a streaming TV. The data type is Boolean (bool).

Example of the Streaming variable data.

StreamingTV
No
Yes

39. StreamingMovies

The StreamingMovies variable indicates whether the customer has streaming movies. The data type is Boolean (bool).

Example of the Streaming variable data.

StreamingMovies
Yes
No

40. PaperlessBilling

The PaperlessBilling variable indicates whether the customer has paperless billing. The data type is Boolean (bool).

Example of the Paperless Billing variable data.

PaperlessBilling	
Yes	
No	

41. PaymentMethod

The PaymentMethod variable indicates the customer's payment method. The data type is a string (str).

Example of the PaymentMethod variable data.

PaymentMethod
Credit Card (automatic)
Bank Transfer(automatic)
Mailed Check
Electronic Check

42. Tenure

The Tenure variable indicates the number of months the customer has stayed with the provider. The data has missing values represented by 'NA.' The data type is float.

Example of the Tenure variable data.

Tenure
6.795513
1.156681
15.75414
17.08723
1.670972
7.000994
13.23677
NA

43. MonthlyCharge

The MonthlyCharge variable represents the amount charged to the customer monthly. This value reflects an average per customer. The data type is float.

Example of the MonthlyCharge variable data.

MonthlyCharge
171.4498
242.948
159.4404
120.2495
150.7612

44. Bandwidth_GB_Year

The Bandwidth_GB_Year represents the average amount of data the customer uses in GB in a year. The data has missing values represented by 'NA.' The data type is float

Example of the Bandwidth_GB_Year variable data.

Bandwidth_GB_Year
904.5361102
800.9827661
2054.706961
2164.579412
1948.694497
NA
1840.014467

45. Item1

The Item1 variable represents customers' timely response ratings where 1 is most important and 8 is least important. The data type is integer (int).

Example of the Item1 variable data.

item1
1
2
3
4
5
6
7

46. Item2

The Item2 variable represents customers' timely fixes ratings where 1 is most important, and 8 is least important. The data type is integer (int).

Example of the Item2 variable data.

item2
1
2
3
4
5
6
7

47. Item3

The Item3 variable represents customers timely replacement ratings where 1 is most important and 8 is least important.

The data type is integer (int).

Example of the Item3 variable data.

item3
1
2
3
4
5
6
7
8

48. Item4

The Item4 variable represents customers' reliability ratings where 1 is most important and 8 is least important.

The data type is integer(int).

Example of the Item4 variable data.

item4
1
2
3
4
5
6
7

49. Item5

The Item5 variable represents customer options ratings where 1 is most important, and 8 is least important. The data type is integer (int).

Example of the Item5 variable data.

item5
1
2
3
4
5
6
7

50. Item6

    The Item 6 variable represents customers' respectful response ratings where 1 is most important and 8 is least important.

    The data type is integer (int).

    Example of the Item6 variable data.

item6
1
2
3
4
5
6
7
8

51. Item7

    The Item7 variable represents customers' Courteous exchange ratings where 1 is most important, and 8 is least important.

    The data type is integer (int).

    Example of the Item7 variable data.

item7
1
2
3
4
5
6
7

52. Item8

The Item8 variable represents customers evidence of active listening ratings where 1 is most important and 8 is least important. The data type is integer (int).

Example of the Item8 variable data.

item8
1
2
3
4
5
6
7
8

B1: Plan to assess the quality of data

According to Stedman, Craig (2022, January), Data cleaning, also referred to as data cleansing or data scrubbing, is the process of fixing incorrect, incomplete, duplicate, or otherwise erroneous data in a data set. It involves identifying data errors and then changing, updating, or removing data to correct them. Data cleansing improves data quality and helps provide more accurate, consistent, and reliable information for decision-making in an organization.

i. I will start by loading the necessary Python Libraries and Packages to assess the data quality.
ii. I will load the data into a data frame using pandas.
iii. I will obtain a summary of the dataframe by using the Python info() method. I will also obtain the statistics of the data frame using the .describe() method.
iv. I will View the Churn_data using head to check for the datatypes for each column and check for the Columns that were loaded with the incorrect datatype. Once incorrect data types have been identified, I will modify the data types for the columns loaded with the incorrect data types.
v. I will Check for missing values and determine how to deal with them depending on the nature of the missingness. Impute the missing data.
vi. I will find Outliers in the data set by utilizing plotting to identify outliers. Grubbs, F.E. states that an outlying observation, or outlier, appears to deviate markedly from other members of the sample in which it occurs.
vii. I will assess the data for duplicate rows by checking for the uniqueness of records using the Customer_id values, reviewing the duplicated rows across all columns, if any, and dropping the complete duplicates. If any duplicates are found, I will assess if they are complete duplicates or partial duplicates and assess the impact of the duplicates before dropping them.
viii. Determine the redundant data and use dimensionality reduction techniques

B2: Justification of approach

In my assessment of the characteristics of the *Churn_raw_data.csv*, the Churn dataset has the following characteristics:

i. The data has boolean, categorical, and numerical data.

ii. The data has values inconsistent with the values of the feature as indicated by "NA" . Hence showing characteristics of missing values.

iii. The data has 10,000 rows and 52 columns, with some unnamed columns.

Based on the characteristics of the data, the approach that I will use to assess the quality of the data will involve;

i. I will use pandas df.dtypes to check the data types for all columns in the data frame.

ii. To assess the missingness of the data on the various variables, I will check using the pandas .isna() function. I will also sum up the missing values to obtain the total missing values per feature and calculate the percentage of missingness. I will use the missingno matrix and bar graphs to check the nature of the missingness.

iii. I will use the Customer_id feature to check for duplicates. I will also use the panda's duplicated method to check for duplicates in the data set.

iv. I will use the Scipy Z-score to assess for any data anomalies and outliers. I will also display the outliers using seaborn boxplots and use Matplotlib histograms to display the distribution of the outliers.

B3: Justification of tools

The programming language selected for this data-cleaning process is Python. Python is an open-source general-purpose programming language with many packages that can be used and shared with others.

I utilized Python Programming Language because of its flexibility and extensive library collections, which are valuable for analytics and performing complex calculations. The code will be created and run through Jupyter Notebook since Jupyter allows for interactive and exploratory coding, live data visualization and analysis, and its dependence on a web browser. Python version 3.12.2 for Windows will be utilized.

The Python libraries and packages that will be imported and utilized in the cleaning process include:

i. Pandas

   Pandas will be used to import and export datasets and explore, manipulate, clean, and analyze data sets.

ii. NumPy

   NumPy will be used for working with arrays and checking data missingness.

iii. Matplotlib

   Matplotlib will be used to create high-quality visualizations and graphs.

iv. Missingno( )

   Missingno( ) will be utilized for visualizing and analyzing missing data patterns in the dataset.

v. Seaborn

   Seaborn will be utilized to create box plots that will show if there are any outliers.

vi. Sklearn

   Sklearn will be utilized in dimensionality reduction.

   Sklearn will be used to impute the missingness in categorical variables.

Sklearn is utilized in the imputation of PCAS

vii. Scipy

Scipy will be used for the computation of outliers.

C1: Cleaning findings

i. Data type constraint

In assessing the data quality, I reviewed the data type constraint and found categorical data loaded as numeric (Float64 and int64). This includes data for the features Zip variable data imported as int64, Lat variable data imported as float, and Lng variable data imported as float. Because of the nature of this data, we cannot be able to compute statistical measures.

```
#Assessing datatype constraint.
Churn_data_df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 52 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unnamed: 0            10000 non-null  int64
 1   CaseOrder             10000 non-null  int64
 2   Customer_id           10000 non-null  object
 3   Interaction           10000 non-null  object
 4   City                  10000 non-null  object
 5   State                 10000 non-null  object
 6   County                10000 non-null  object
 7   Zip                   10000 non-null  int64
 8   Lat                   10000 non-null  float64
 9   Lng                   10000 non-null  float64
```

ii. Data Missingness

From the Churn dataset, there was a data quality issue of missingness found in the data. The missing data was completely random. The percentage of missingness was quite high, as demonstrated below. Buuren and Stef van (2018) state that if the probability of being missing is the same for all cases, then the data are said to be missing completely at random (MCAR). This effectively implies that the causes of the missing data are unrelated to the data.

Variables with missing values include:

1. Children.

   The Children variable has 2495 missing variables, which account for 24.95% of the values. The Missingness is completely at random.

2. Age

   The Age variable has 2475 missing variables, which account for 24.75% of the values.

3. Income

   The Income variable has 2490 missing variables, which account for 24.90% of the values.

4. Techie

   The Techie variable has 2477 missing variables, which account for 24.77% of the values.

5. Phone

   The Phone variable has 1026 missing variables, which account for 10.26% of the values.

6. TechSupport

   The TechSupport variable has 991 missing variables, which account for 9.91% of the values.

7. Tenure

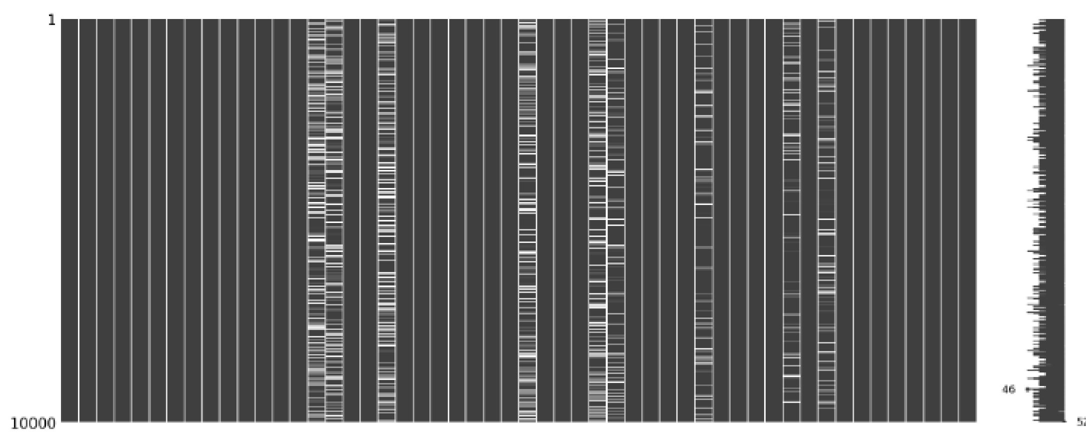
   The Tenure variable has 931 missing variables, which account for 9.31% of the values.

8. Bandwidth_GB_Year

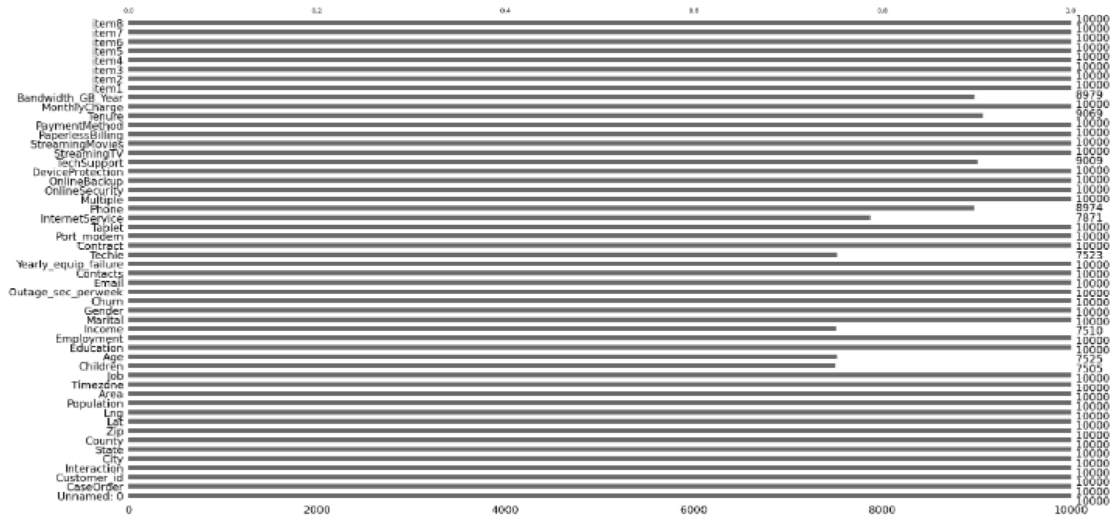
   The Bandwidth_GB_Year variable has 1021 missing variables, which account for 10.21% of the values.

overview of the missing data from the dataset

```
In [10]:  #Displaying the missingness using a matrix
          msno.matrix(Churn_data_df)
Out[10]:  <Axes: >
```



```
In [11]: #Displaying the missingness using a bar
         msno.bar(Churn_data_df)
Out[11]: <Axes: >
```



```
Out[10]: Unnamed: 0                0
         CaseOrder                 0
         Customer_id               0
         Interaction               0
         City                      0
         State                     0
         County                    0
         Zip                       0
         Lat                       0
         Lng                       0
         Population                0
         Area                      0
         Timezone                  0
         Job                       0
         Children               2495
         Age                    2475
         Education                 0
         Employment                0
         Income                 2490
         Marital                   0
         Gender                    0
         Churn                     0
         Outage_sec_perweek        0
         Email                     0
         Contacts                  0
         Yearly_equip_failure      0
         Techie                 2477
         Contract                  0
         Port_modem                0
         Tablet                    0
         InternetService           0
         Phone                  1026
         Multiple                  0
         OnlineSecurity            0
         OnlineBackup              0
         DeviceProtection          0
         TechSupport             991
         StreamingTV               0
         StreamingMovies           0
         PaperlessBilling          0
         PaymentMethod             0
         Tenure                  931
         MonthlyCharge             0
         Bandwidth_GB_Year      1021
         item1                     0
         item2                     0
         item3                     0
         item4                     0
         item5                     0
         item6                     0
         item7                     0
         item8                     0
         dtype: int64
```

iii. Data uniqueness

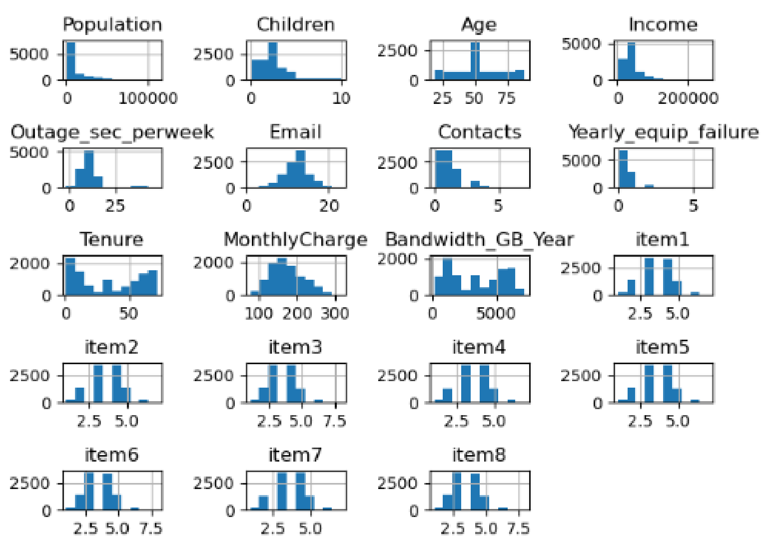
Data uniqueness was validated as part of the data cleaning, and no partial or complete duplicate records existed.

iv. Outliers

As part of the data cleaning, validation for outliers and anomalies was performed, and outliers were found in the features of Population, Children, Income, Outage_sec_perweek, Email, Contacts, Yearly_equip_failure, MonthlyCharge, item1, item2, item3, item4, item5, item6, item7and item8. There were no outliers in the features of Age, Tenure and Bandwidth_GB_Year.

In [36]:
```
# Utilize Histograms for anomaly/outliers detection and feature visualization

Churn_data_df[['Population', 'Children', 'Age','Income' , 'Outage_sec_perweek', 'Email', 'Contacts', 'Yearly_equip_failure',
plt.savefig ('churn_pyplot.jpg')
plt.tight_layout()
```



C2: Justification of mitigation methods

i. Data type constraint

The Categorical data below was converted from numerical data types to categorical data types because, by the nature of the variables, they cannot be used to impute statistical measures.

1. Zip variable data was converted from integer to string.
2. Lat variable data was converted from float to string.
3. Lat variable data was converted from float to string.

ii. Data Missingness

Data missingness was addressed by imputing the missing values for the numerical data using the mean value. This method was chosen because the missing values were many, so the data could not be dropped; hence, I opted to use statistical measure (mean)to impute the missing values. On the other hand, the categorical data was not utilized for the purpose of this analysis. However, the missingness was imputed using the most frequent value (mode). This is because dropping the missing data would impact the size of the data set.

1. Children.

   The missingness in the Children variable was imputed using the mean.

2. Age

   The missingness in the Age variable was imputed using the mean.

3. Income

   The missingness in the Income variable was imputed using the mean.

4. Techie

   The missingness in the Techie variable was imputed using the mode (most common value).

5. Phone

   The missingness in the Phone variable was imputed using the mode (most common value).

6. TechSupport

   The missingness in the TechSupport variable was imputed using the mode (most common value).

7. Tenure

   The missingness in the Tenure variable was imputed using the mean.

8. Bandwidth_GB_Year

   The missingness in the Bandwidth_GB_Year variable was imputed using the mean.

iii. Data uniqueness

No duplicates were found in the data; hence, there was no mitigation on the data.

iv. Outliers

The Outliers in the dataset variables Population, Children, Income, Outage_sec_perweek, Email, Contacts, Yearly_equip_failure, MonthlyCharge, item1, item2, item3, item4, item5, item6, item7 and item8 were not removed from the datasets due to

1. Lack of domain knowledge.
2. Lack of understanding of the data collection process.
3. Outliers can provide valuable insights into the data.
4. Lack of a clear threshold value to use for the removal of the data with outliers.

C3: Summary of the outcome from the implementation of *each* data-cleaning step.

Below is the summarized outcome of each data-cleaning step.

i. Categorical loaded as numerical data in the Zip, Lat, and Lng features was converted from numerical type to Categorical data types.

ii. The missingness of data in several features with numerical data type was imputed using mean values, whereas the missingness for categorical data was imputed using the most frequent or mode.

iii. No duplicates were found; hence, there was no uniqueness constraint.

iv. Various data points in the variables Population, Children, Income, Outage_sec_perweek, Email, Contacts, Yearly_equip_failure, MonthlyCharge, item1, item2, item3, item4, item5, item6, item7and item8 had outliers as identified via the zscores, histogram, and bar graph; however, the outliers could not be explained due to a lack of domain knowledge. The outliers were retained in the data set.

C4: Limitations

The limitations that were experienced in the data cleaning process include:

i. We used statistical measures (mean and mode) to impute numerical and categorical data missingness. Dropping the missing values would have significantly reduced the sample size. On the other hand, using the mean and mode introduced biases in the data and affected the distribution of the data.
ii. I also lacked a clear understanding of the data due to a lack of domain knowledge and expertise.
iii. Outliers were not removed due to the lack of knowledge of the domain or even the data collection process. Moreover, there was no defined threshold value for removing the outliers.

C5: Impact of limitations

The impact of the limitations in the data cleaning process includes:

i. Imputing the missing data using statistical measures assumes that the missing values take on the shape of the statistical measures, hence introducing a bias in the data. on the other hand, imputation may provide a path to move forward and give decision-makers an analysis that will help them make the right decisions.

ii. Lack of domain knowledge and the data collection process that was used limited how the anomalies were mitigated.

iii. Outliers would cause the analysis to miss significant results or distort the real findings of the analysis.

D1:Mitigation code for the data quality issues

Comments

Hello, per my discussion with Dr. Keiona, we agreed that I attach the executable code, the clean CSV churn data file as part of the attachments, and the link for the Panopto recording. I would also like to bring to your attention that the file path that is on the executable code is on my local C drive; therefore, if you execute the code as is, then you will get an error, and hence the need to update the file path with the drive that you're working on. For ease of reference, I have defined the path as variables below

#Defining the input and output file paths and holding them in variables.

Input_file_path= 'C:/Users/beatr/Downloads/churn_raw_data.csv'

Output_file_path ='C:/Users/beatr/Downloads/churn_clean_data.csv'

thank you.

Jupiter notebook executable file

BeatriceMwangi_D2
06_PerfomanceAsses

D2: Clean data

Cleaned Churn data

churn_clean_data.csv

E1:Principal Components

According to Jaadi, Zakaria (2024, February 23), Principal component analysis (PCA) is a dimensionality reduction and machine learning method that is used to simplify a large dataset into a smaller set while still maintaining significant patterns and trends.

Jaadi, Zakaria (2024, February 23) defines the principal components as the new variables that are constructed as linear combinations or mixtures of the initial variable.

From the principal component analysis below, 51% of the variance is explained by 6 Components. And 90% of the variance is explained by 14 components. The first principal component explains 15% of the variance. Based on the matrix below, items 1, 2, and 6 are the most important components, meaning that customers in the telecommunication industry value timely responses, timely fixes, and respectful responses when determining whether to switch providers in the first Principal component.

	PCA1	PCA2	PCA3	PCA4	PCA5	PCA6	PCA7	PCA8	PCA9
Population	-0.002116	0.000523	0.015349	-0.045365	-0.303168	-0.385048	0.049662	0.695744	0.415015
Children	0.001156	0.001017	0.010821	-0.001467	0.569318	-0.197519	-0.081648	0.187028	-0.494931
Age	0.004999	-0.013349	-0.017168	-0.043357	-0.403609	0.466462	0.174021	0.138800	-0.364653
Income	-0.000895	0.007842	0.025254	-0.002042	0.213787	0.291447	-0.712348	0.448086	-0.000909
Outage_sec_perweek	-0.013144	0.018367	-0.047482	0.703595	0.035817	-0.010298	-0.013264	0.061760	0.030818
Email	0.008640	-0.021059	-0.004266	0.053667	-0.296917	-0.558577	-0.000758	0.126659	-0.606219
Contacts	-0.008584	0.003706	-0.010669	-0.007416	-0.447096	0.294470	-0.250238	0.083423	-0.250321
Yearly_equip_failure	-0.007711	0.016091	0.007008	0.059135	0.276463	0.329158	0.620419	0.477958	-0.102967
Tenure	-0.010877	0.701077	-0.071115	-0.058352	-0.019944	-0.005494	0.003961	-0.001806	-0.011235
MonthlyCharge	-0.000540	0.043675	-0.024571	0.697085	-0.083053	0.030220	-0.032908	-0.057175	0.020128
Bandwidth_GB_Year	-0.012700	0.702809	-0.073361	-0.010082	0.003980	-0.017034	-0.004158	-0.003263	-0.009114
item1	0.458788	0.032053	0.280142	0.031659	0.006166	0.006133	0.017106	-0.000754	-0.001130
item2	0.434074	0.042978	0.281346	0.018576	-0.009353	0.017472	-0.001691	-0.010845	-0.009467
item3	0.400745	0.034836	0.280786	-0.010542	-0.006822	-0.024290	0.020997	-0.025823	0.005827
item4	0.145679	-0.049876	-0.567103	-0.031501	-0.000794	-0.001123	0.016038	-0.012034	0.020591
item5	-0.175538	0.065834	0.585933	0.026388	-0.034716	0.016285	0.005485	-0.017244	0.005444
item6	0.405058	-0.011897	-0.183620	0.005088	-0.002970	-0.003886	-0.002052	0.022865	-0.009946
item7	0.358277	-0.003472	-0.180784	-0.032480	0.021392	0.012288	-0.055150	0.006840	-0.036084
item8	0.308763	-0.016869	-0.132055	0.030267	-0.020877	0.020629	0.015612	0.023986	0.074333

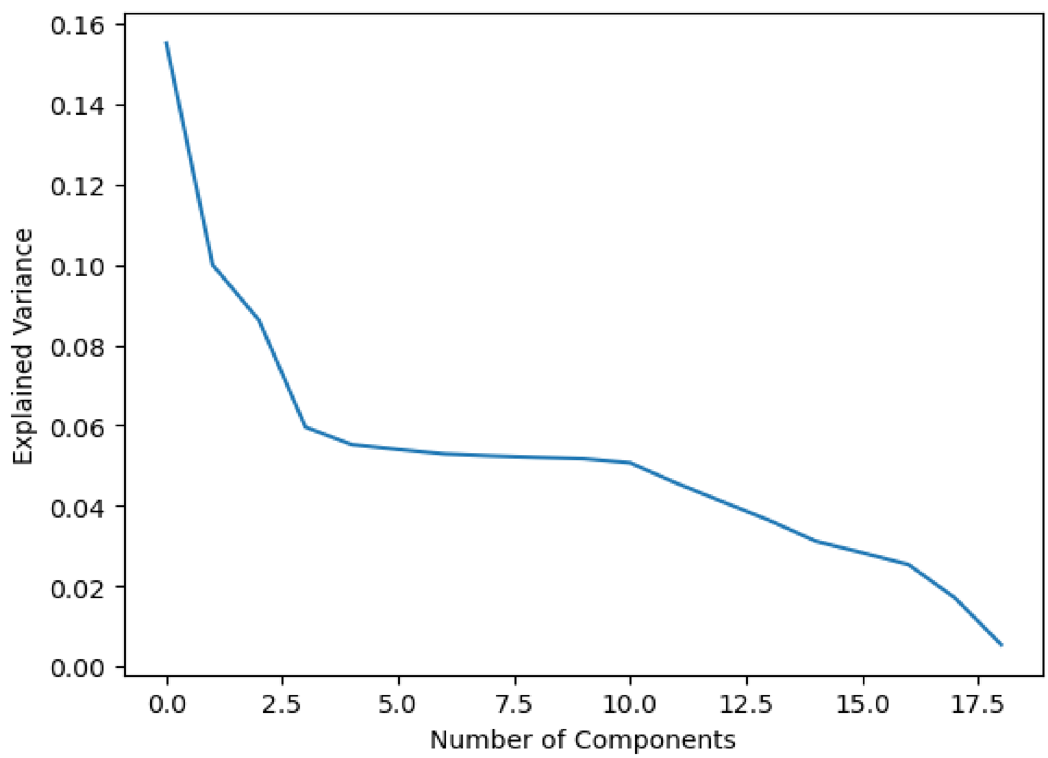
	PCA10	PCA11	PCA12	PCA13	PCA14	PCA15	PCA16	PCA17	PCA18	PCA19
Population	0.006749	-0.306589	-0.015703	-0.061065	0.020910	-0.016697	0.001026	-0.004950	-0.002364	-0.000863
Children	0.114080	-0.577823	0.008129	0.027453	-0.031503	-0.015521	0.013451	0.023687	-0.006177	-0.020305
Age	-0.551945	-0.336491	0.108479	-0.062576	0.010630	-0.001214	-0.012073	0.008674	0.015987	0.021706
Income	-0.229464	0.316054	-0.063511	0.012061	-0.059503	-0.002897	0.000681	0.011446	0.004777	0.001507
Outage_sec_perweek	0.026937	0.040781	0.692017	-0.117783	0.012292	-0.012063	-0.016843	0.010549	-0.004330	0.000677
Email	-0.110651	0.439943	-0.041248	0.066355	-0.018311	-0.016279	0.007237	-0.016981	0.001104	0.005521
Contacts	0.751132	-0.111800	0.016482	0.038350	-0.036401	-0.003701	-0.025898	0.020960	-0.000826	-0.002707
Yearly_equip_failure	0.209511	0.356868	-0.120658	0.028985	0.004639	-0.015081	-0.000820	0.006872	-0.021031	-0.002307
Tenure	-0.019719	0.012460	0.038123	0.000748	-0.010115	0.007694	-0.012192	0.006401	0.004737	-0.704958
MonthlyCharge	-0.061830	-0.142401	-0.685107	0.047792	0.009663	-0.014504	0.000955	0.021254	-0.013215	-0.048020
Bandwidth_GB_Year	0.001516	-0.007802	-0.011405	0.013278	0.002577	0.003235	-0.001554	-0.008216	0.007631	0.706741
item1	0.022180	-0.000582	-0.007400	-0.071436	-0.118202	-0.047451	0.024122	-0.239332	0.793025	-0.003379
item2	0.007436	-0.013379	0.000148	-0.111554	-0.169824	-0.065941	0.070243	-0.591735	-0.573297	-0.002856
item3	-0.007657	0.018692	-0.010986	-0.177187	-0.248053	-0.146210	-0.393143	0.675410	-0.177947	0.014944
item4	0.005188	0.009003	-0.023497	-0.179323	-0.478397	-0.443616	0.431879	0.086898	0.017750	0.001805
item5	-0.005016	-0.004536	0.042283	0.130084	0.060023	-0.208780	0.694728	0.261918	-0.042523	-0.002847
item6	0.014018	0.007185	-0.004684	-0.061335	0.064417	0.758162	0.403427	0.227103	-0.063804	0.001647
item7	0.028047	0.023770	-0.024127	-0.162500	0.807032	-0.377598	0.068080	0.065358	-0.041400	-0.006913
item8	-0.052199	-0.046896	0.119968	0.916635	-0.024962	-0.113080	-0.043370	0.046404	-0.042772	-0.003353

```
PCA1 0.15515780194791706
PCA2 0.25515591960729944
PCA3 0.3413139561502836
PCA4 0.4008919559767109
PCA5 0.45610234899911634
PCA6 0.5101070282335861
PCA7 0.5629862631890432
PCA8 0.6153808335036344
PCA9 0.6673970390206292
PCA10 0.7190924248170667
PCA11 0.7697686080893325
PCA12 0.8153674280762457
PCA13 0.8563183123079597
PCA14 0.8926765630439344
PCA15 0.9238480656116562
PCA16 0.9521496822575292
PCA17 0.9775050323758175
PCA18 0.994591149616668
PCA19 0.9999999999999999
```

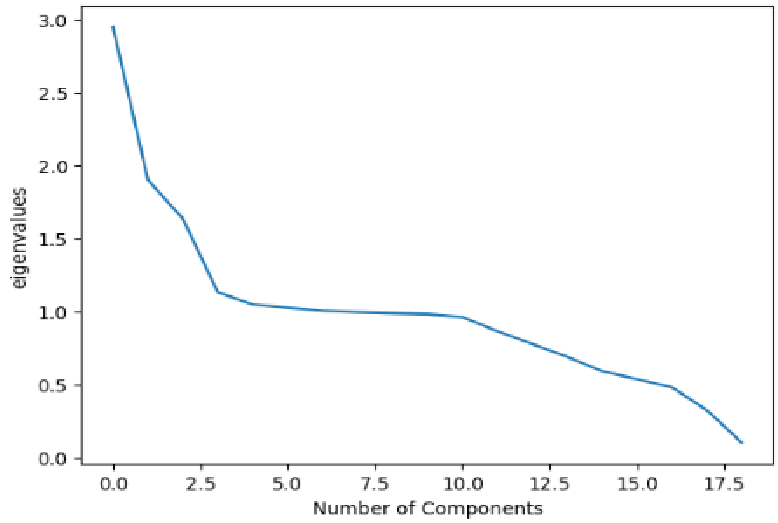
E2:Criteria used

Mangale, Sanchita (2020, August 28) explain that a scree plot is a common method for determining the number of principal components to be retained in a graphical representation. A scree plot is a simple line segment plot showing each PC's eigenvalues. The first principal components from the scree plot below explain the highest variability as explained by the elbow. After the third principal component, where the curve starts to flatten, there is no real benefit of using more than three components because all the components beyond that.

```
# Create the scree plot
plt.plot(pca.explained_variance_ratio_)
plt.xlabel('Number of Components')
plt.ylabel('Explained Variance')
plt.show();
```



```
# Plot the eigenvalues
plt.plot(eigenvalues)
plt.xlabel('Number of Components')
plt.ylabel('eigenvalues')
plt.show();
```



E3: Benefits of Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction method that offers several benefits for organizations. According to Jain, Sandeep (2023, December 06), the benefits of PCA include:

i. Dimensionality reduction.

PCA will prevent overfitting of data especially when dealing with high dimensional data. Overfitting occurs when there are too many features in a dataset, leading to a model that performs well on the training data but poorly on unseen data. PCA helps mitigate this issue by reducing the dimensionality of the dataset, which can prevent overfitting.

ii. Feature Selection

Principal component analysis can be used for selecting the most important feature in a dataset.

iii. Multicollinearity

PCA also removes Correlated Features. Correlation can make it challenging to interpret the effect of individual features on the predicted outcome. PCA simplifies the model interpretation by reducing correlation in a dataset.

iv. Data Compression

By using PCA, Machine Learning Algorithms speed can be increased by using the principal components derived from Principal Component Analysis instead of all the original features, this improves efficiency and reduces training time.

v. Data visualization:

Principal Component Analysis modifies the data into fewer features, making it easier to visualize and explore patterns.

vi. Outlier detection:

Principal component analysis can be used for outlier detection by looking at the data points that are far from the other points in the principal components' space.

F: video

Beatrice_Mwangi_D206_DataCleaning_PerfomanceAssessment (panopto.com)

G: sources of third-party code

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